

Name: ARYAN SINGH SIKARWAR

Registration No: 24BCE0403

Probability and Statistics Lab (DA-3)

1. A pharmaceutical company reports that 5% of tablets produced are defective. A batch of 20 tablets is selected randomly for inspection. Solve using Binomial distribution to find the probability that exactly 2 tablets are defective.

CODE:

The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for a binomial distribution problem. The code defines variables for the number of trials (n=20), probability of defect (p=0.05), and the number of defective tablets (x=2). It then calculates the probability using the `dbinom` function and prints the result.
- Console:** Shows the execution of the code. It displays the error message "Error: object 'probability' not found" because the variable `probability` was not assigned a value before being used in the `print` statement. The corrected code is shown below the error.
- Environment:** Lists the objects in the global environment, including `model`, `new_data`, and `spearman_test`.
- Files:** Shows the file explorer with a list of files in the `dhcnsdfb` directory, including `.RData`, `.Rhistory`, `dhcnsdfb.Rproj`, and `Untitled.R`.

```
# Given values
n <- 20 # number of trials
p <- 0.05 # probability of defect
x <- 2 # exactly 2 defective tablets

# Binomial probability
probability <- dbinom(x, size = n, prob = p)

# Print result
probability
```

Console Output:

```
> # Given values
> n <- 20 # number of trials
> p <- 0.05 # probability of defect
> x <- 2 # exactly 2 defective tablets
> 
> # Binomial probability
> probability <- dbinom(x, size = n, prob = p)
> 
> # Print result
> probability
[1] 0.1886768
>
```

Que 1 (code part):-

Given values

```
n <- 20 # number of trials p <- 0.05 # probability of defect x <- 2 # exactly 2 defective tablets
```

Binomial probability

```
probability <- dbinom(x, size = n, prob = p)
```

Print result

```
probability
```

QUE-2

1. in an online order processing system, the probability that an order is successfully completed without error is 0.92. If 10 orders are processed independently, solve using Binomial distribution to determine the probability that at least 8 orders are completed successfully.

CODE:-

The screenshot displays the RStudio environment with the following components:

- Script Editor:** Contains R code for a binomial probability calculation.

```
1 # Given values
2 n <- 10 # number of orders
3 p <- 0.92 # probability of success
4
5 # Probability that at least 8 orders are successful
6 probability <- pbinom(7, size = n, prob = p, lower.tail = FALSE)
7
8 # Print result
9 probability
10
```
- Console:** Shows the execution of the script, including the initial probability value and the final result after the binomial calculation.

```
> probability
[1] 0.1886768
> # Given values
> n <- 10 # number of orders
> p <- 0.92 # probability of success
>
> # Probability that at least 8 orders are successful
> probability <- pbinom(7, size = n, prob = p, lower.tail = FALSE)
>
> # Print result
> probability
[1] 0.9599246
> # Print result
> probability
[1] 0.9599246
>
```
- Environment Pane:** Lists objects in the Global Environment.

Object	Class	Value
model	List of 12	
new_data	1 obs. of 2 variables	
spearman_test	List of 8	
d	num [1:10]	-1 1 1 0 1 -1 -1 0 1
d2	num [1:10]	1 1 1 0 1 1 1 1 0 1
intercept	Named num	0.794
n	10	
p	0.92	
predicted_sales	Named num	532
predicted_value	Named num	27.1
probability	0.959924580316495	
rank_X	num [1:10]	7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10]	8 3 9 2 7 4 6 10 1 5
rho_manual	0.951515151515152	
slope	Named num	0.142
spearman_corr	0.951515151515151	
x	2	
- Files Pane:** Shows the project structure.

Name	Size	Modified
..		
.RData	4.7 KB	Feb 11, 2026, 9:39 PM
.Rhistory	2.2 KB	Feb 11, 2026, 9:39 PM
dhcnsdfb.Rproj	233 B	Mar 4, 2026, 3:16 PM
Untitled.R	224 B	Mar 4, 2026, 3:19 PM

Que -2(code part):-

Given values

`n <- 10` # number of orders `p <- 0.92` # probability of success

Probability that at least 8 orders are successful

`probability <- pbinom(7, size = n, prob = p, lower.tail = FALSE)`

Print result

probability

Que-3:-

1. A help desk receives an average of 3 service requests every hour. Solve using Poisson distribution to find the probability that exactly 5 requests are received in a particular hour.

CODE:-

The screenshot displays the RStudio environment with the following components:

- Script Editor:** Contains R code for calculating Poisson probability.
- Console:** Shows the execution of the script, resulting in the probability value 0.1008188.
- Environment Pane:** Lists variables created in the global environment, including 'model', 'new_data', 'spearman_test', and various model coefficients.

```
1 # Given values
2 lambda <- 3 # average requests per hour
3 x <- 5 # exactly 5 requests
4
5 # Poisson probability
6 probability <- dpois(x, lambda)
7
8 # Print result
9 probability
```

Console Output:

```
> probability
[1] 0.9599246
> # Print result
> probability
[1] 0.9599246
> # Given values
> lambda <- 3 # average requests per hour
> x <- 5 # exactly 5 requests
>
> # Poisson probability
> probability <- dpois(x, lambda)
>
> # Print result
> probability
[1] 0.1008188
>
```

Environment Variables:

Variable	Value
model	List of 12
new_data	1 obs. of 2 variables
spearman_test	List of 8
d	num [1:10] -1 1 1 0 1 -1 -1 0 1
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1
intercept	Named num 0.794
lambda	3
n	10
p	0.92
predicted_sales	Named num 532
predicted_value	Named num 27.1
probability	0.100818813444924
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5
rho_manual	0.951515151515152
slope	Named num 0.142
spe	slope (numeric, 280 bytes) 51515151515151
x	5

Que-3(code part):-

Given values

```
lambda <- 3 # average requests per hour x <- 5 # exactly 5 requests
```

Poisson probability

```
probability <- dpois(x, lambda)
```

Print result

```
probability
```

Que-4:-

1. A railway reservation counter observes that an average of 2 customers arrive every 10 minutes. Solve using Poisson distribution to determine the probability that fewer than 2 customers arrive during a 10-minute interval.

Code:-

The screenshot displays the RStudio environment with the following components:

- Script Editor (Untitled1*):** Contains R code for a Poisson probability calculation.


```

1 # Given value
2 lambda <- 2 # average customers in 10 minutes
3
4 # Probability that fewer than 2 customers arrive
5 probability <- ppois(1, lambda)
6
7 probability
      
```
- Console:** Shows the execution of the code, resulting in the probability value 0.1008188.


```

> # Poisson probability
> probability <- dpois(x, lambda)
>
> # Print result
> probability
[1] 0.1008188
> # Given value
> lambda <- 2 # average customers in 10 minutes
>
> # Probability that fewer than 2 customers arrive
> probability <- ppois(1, lambda)
>
> probability
[1] 0.4060058
      
```
- Environment Pane:** Lists variables in the Global Environment.

Variable	Value
model	List of 12
new_data	1 obs. of 2 variables
spearman_test	List of 8
d	num [1:10] -1 1 1 0 1 -1 -1 -1 0 1
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1
intercept	Named num 0.794
lambda	2
n	10
p	0.92
predicted_sales	Named num 532
predicted_value	Named num 27.1
probability	0.406005849709838
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5
rho_manual	0.951515151515152
slope	num [1:10] 8 3 9 2 7 4 6 10 1 5
spearman_corr	0.951515151515151
- Files Pane:** Shows the project structure with files like .RData, .Rhistory, dhcnscdfb.Rproj, and Untitled.R.

Question-4(code):-

Given value

lambda <- 2 # average customers in 10 minutes

Probability that fewer than 2 customers arrive

probability <- ppois(1, lambda)

Probability

Que-5:-

1. The lifetime of LED bulbs produced in a factory follows a normal distribution with mean 1200 hours and standard deviation 150 hours. Solve using Normal distribution to find the probability that a bulb lasts more than 1400 hours.

Code:-

```
1 # Given values
2 mean <- 1200
3 sd <- 150
4 x <- 1400
5
6 # Probability that bulb lasts more than 1400 hours
7 probability <- pnorm(x, mean = mean, sd = sd, lower.tail = FALSE)
8
9 probability
```

Console Output:

```
> # Probability that fewer than 2 customers arrive
> probability <- ppois(1, lambda)
>
> probability
[1] 0.4060058
> # Given values
> mean <- 1200
> sd <- 150
> x <- 1400
>
> # Probability that bulb lasts more than 1400 hours
> probability <- pnorm(x, mean = mean, sd = sd, lower.tail = FALSE)
>
> probability
[1] 0.09121122
```

Environment Pane:

Object	Class	Value
spearman_test	List of 8	
d	num [1:10]	-1 1 1 0 1 -1 -1 0 1
d2	num [1:10]	1 1 1 0 1 1 1 1 0 1
intercept	Named num	0.794
lambda		2
mean		1200
n		10
p		0.92
predicted_sales	Named num	532
predicted_value	Named num	27.1
probability		0.0912112197258678
rank_X	num [1:10]	7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10]	8 3 9 2 7 4 6 10 1 5
rho_manual		0.951515151515152
sd		150
slope	Named num	0.142
spearman_corr		0.951515151515151
x		1400

Code-part:-

Given values

```
mean <- 1200 sd <- 150 x <- 1400
```

Probability that bulb lasts more than 1400 hours

```
probability <- pnorm(x, mean = mean, sd = sd, lower.tail = FALSE)
```

Probability

Que-6:-

1. The heights of students in a college are normally distributed with mean 165 cm and standard deviation 8 cm. Solve using Normal distribution to determine the probability that a randomly selected student has height between 160 cm and 175 cm.

Code:-

The screenshot shows the RStudio interface with the following components:

- Script Editor (Untitled1.R):**

```

1 # Given values
2 mean <- 165
3 sd <- 8
4
5 # Probability between 160 and 175
6 probability <- pnorm(175, mean, sd) - pnorm(160, mean, sd)
7
8 probability

```
- Console:**

```

> # Probability that bulb lasts more than 1400 hours
> probability <- pnorm(x, mean = mean, sd = sd, lower.tail = FALSE)
>
> probability
[1] 0.09121122
> # Given values
> mean <- 165
> sd <- 8
>
> # Probability between 160 and 175
> probability <- pnorm(175, mean, sd) - pnorm(160, mean, sd)
>
> probability
[1] 0.6283647
>

```
- Environment:**
 - spearman_test**: List of 8

Values	
d	num [1:10] -1 1 1 0 1 -1 -1 0 1
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1
intercept	Named num 0.794
lambda	2
mean	165
n	10
p	0.92
predicted_sales	Named num 532
predicted_value	Named num 27.1
probability	0.628364697284444
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5
rho_manual	0.951515151515152
sd	8
slope	Named num 0.142
spearman_corr	0.951515151515151
x	1400
- Files:**
 - Home > dhcnsdfb

Name	Size	Modified
..		
.RData	4.7 KB	Feb 11, 2026, 9:39 PM
.Rhistory	2.2 KB	Feb 11, 2026, 9:39 PM
dhcnsdfb.Rproj	233 B	Mar 4, 2026, 3:16 PM
Untitled.R	224 B	Mar 4, 2026, 3:19 PM

Code-part:-

Given values

```
mean <- 165 sd <- 8
```

Probability between 160 and 175

```
probability <- pnorm(175, mean, sd) - pnorm(160, mean, sd)
```

```
probability
```